Valuation Principles

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Chapter 1

Valuation Methods

Learning Objectives

By the end of this chapter, students will be able to:

1. Distinguish among the six major valuation methods and identify the best use case for each model
2. Define capital assets and contracts, and explain why both require valuation.
3. Be introduced to basic securities and contracts: Stocks, bonds, forwards, futures and options.
4. Calculate dollar returns and percentage returns, and recognize when percentage returns are misleading or undefined.
5. Apply the General Valuation Equation (GVE) to price a single risk-free cash flow and a risky asset.
6. Explain the relationship between expected return and required return in a competitive capital market.
7. Learn how to adjust the GVE to value assets and contracts with different risk measures.

Introduction

Problems in valuation extend far beyond the scope of this text. Most finance students will not be called upon to value antique paintings or used cars, so we may consider it safe to exclude these interesting cases. However, finance professionals are clearly expected to value financial instruments, stocks and bonds being the obvious examples, most of which would have no value were it not for the contractual or otherwise expected cash flows they generate. We call these instruments *capital assets*. As you might expect, capital assets trade in *capital markets*.

In this text, we define our coverage of assets to include both capital assets and physical assets whose values derive from the cash flows they produce, such as properties that produce rental income, but not the intangible benefits they provide to any individual or institution. This does not mean to suggest that intangible benefits have no value. Intangible benefits remind us that valuation models are often incomplete, and we should remember that precious few models can be perfectly complete.

Also, if we limit our valuation scope to capital assets and physical assets that generate cash, it would not suffice. We must also include *contracts* in our valuation models, even though they may require little or no initial cash investment. Why? Because a contract generally becomes either an asset or a liability over time, even if it appeared to be of equal value to both parties with a total value equal to zero when it was originally agreed. For example, if you sign a contract giving you the exclusive right to sell hot dogs at Yankee Stadium, that contract value might rise or fall significantly over its life as the team's fortunes and attendance levels change. The value may even

become negative as the contract's financial obligations exceed the benefits of the contract.

How is value generally computed? The most popular valuation model for cash flows is called "Present Value" or "Net Present Value." These models use forecasted expected future cash flows together with assumed timing and risk adjustments to find an asset's value today. But future cash flows are usually not known with certainty, and the risk adjustments are usually subjective. So what is the value of a model then, if it does not produce a complete and perfectly supportable valuation?

> *"Remember that all models are wrong; the practical question is how wrong do they have to be to not be useful."* (Box and Draper, 1987)

Modelers should humbly keep this aphorism in mind. Present value is not the same as value; it is but one model for valuation. There are many possible models, and any one of them might be most appropriate for a particular purpose at a point in time. This chapter surveys those models, then introduces the single equation that unifies all of them.

1.1 Asset and Contract Types

Assets may be tangible, such as a building or a ship. They may be intangible, such as a corporate investment opportunity, a celebrity's brand, or even a graduate education. Somewhere in between lie capital assets, which are claims to cash flows. An example of a capital asset is a share of stock, which provides its owner with a proportional share of the appreciation and cash flows of an enterprise. The appreciation in price is called the *capital gain*, and corporate payouts to shareholders are called *dividends*.

Investors buy stocks when they think that their value will rise, or that they contribute to the overall quality of an investment portfolio. If a speculator thinks the value of a stock will fall, she can *sell short*, which means she can borrow the stock from a broker, sell it in the stock market, and buy it back at some future date. If the stock's value is lower on the future date, she will make a profit. If the stock's value is higher, she will experience a loss. You might say, "she is betting against the market."

A bond is another example of a capital asset. Bondholders receive periodic interest payments called *coupons*, plus the return of principal at the *maturity date*, in exchange for an initial cash investment. The timing and promised amount of payments is known at the time of investment. Bonds are not always safe, they may default, and their values fluctuate as interest rates change.

A third example is income-producing real estate: owners receive capital gains and rental income over time while paying operating expenses.

Assets can also include contracts, which are often classified as *contingent assets*. Unlike capital assets, contracts are two-sided agreements: what is an asset for one party is a liability for the other. For example, a call option gives the buyer the right to purchase a stock at a fixed price for some period of time. That right has potential value for the buyer, but it creates a corresponding

obligation for the seller. Contracts typically start with zero net value but can evolve significantly as market conditions change. Put options resemble calls, except that the owner of the put option has the right to sell an asset at a fixed price for some period of time.

Forward contracts provide a way for investors to take the risk of an asset without any up-front payment. If someone "goes long" in a forward position, they will irrevocably agree to buy the asset at a future date at a fixed price. If they "go short", then they are irrevocably agreeing to sell. When they are "long", they make money if the underlying asset value increases and lose money if it decreases, even though they don't legally own the asset until the contracted date. If they are "short" they have exactly the opposite profit and loss (P/L or P&L) profile. *Futures contracts* are forward contracts traded on futures exchanges, which will be discussed in depth in a later chapter.

A key legal concept affecting asset value is *limited liability.* If a corporation fails, its creditors may not reclaim money from shareholders beyond their original equity investment. This asymmetry, that shareholders capture the upside but lose at most their stake, has significant valuation implications we develop throughout the text.

The total return on any capital asset has two components. The cash flow yield includes dividends on a stock, coupons on a bond, or rent on a property. The capital gain is the change in the asset's market value over the holding period. Both components matter. The table below organizes the major instrument types we will study.

Instrument	Category	Primary Cash Flows	Counterparty Position
Stock	Capital asset	Dividends + capital gains	None (limited liability)
Bond (investor)	Capital asset	Coupons + principal	Issuer's liability
Real estate	Capital asset	Rent + appreciation	None
Mortgage (borrower)	Liability	Interest + principal owed	Lender's asset
Forward / futures	Contract	Gain or loss vs. agreed price	Symmetric and opposite
Call option (buyer)	Contingent asset	Payoff if price $>$ strike	Writer's contingent liability
Put option (buyer)	Contingent asset	Payoff if price $<$ strike	Writer's contingent liability

A first look at derivatives. The last three rows of the table above show three derivative instruments: forwards/futures, call options, and put options. A derivative is a contract whose value comes from an underlying asset (a stock, bond, commodity, or index) rather than from any cash flow it generates on its own. Because the text of this book leans heavily on derivatives for example, it is worth pausing to review these contracts.

A forward contract is a binding agreement, struck today, to exchange a fixed quantity of the underlying asset at a fixed price (called the *delivery price*) on a fixed future date. Generally, no cash changes hands at inception. The forward buyer of the contract (the *long*) profits dollar for dollar when the underlying ends above the delivery price and loses dollar for dollar when it ends below. The seller (the *short*) has the opposite payoff. A futures contract is an exchange-traded, standardized forward; the economics are identical but the cash flows are settled daily rather than at maturity.

An option gives its buyer the right, but not the obligation, to buy or sell the underlying at a fixed price called the *strike price*, denoted X. A call gives the right to buy at price X; a put gives the right to sell at X. The buyer pays a premium up front to acquire that right. At expiration, the holder of a call earns $max(S - X, 0)$: if the underlying S ends above X they exercise and pocket the difference; otherwise they walk away. A put pays $max(X - S, 0)$ by the same logic. The seller (the *writer*) collects the premium and assumes the opposite liability. This asymmetric payoff is what separates options from linear instruments like stocks or forwards, and it is the reason a separate chapter is devoted to them later in the book.

A note on bonds and fixed income. Bonds, treasury bills(T-bills), mortgages, and similar instruments are commonly referred to as *fixed income securities*. The label reflects the structure of the cash flows: a stream of contractually fixed payments at known dates (a periodic coupon and the principal at maturity, or amortizing payments in the case of a mortgage), discounted back to today. Because the size and timing of the cash flows are typically known up front, the entire valuation question reduces to choosing the right discount rate. That discount rate, the *yield*, changes over time as inflation expectations, credit risk, and central-bank policies change. The price of a fixed income security is the present value of its known cash flows at the current yield, which is why yields and prices move in opposite direction. We return to this in detail in the bond chapter.

1.2 Valuation Models

There are many possible models of valuation, and any one of them might be most appropriate for a particular purpose at a point in time. The table below describes six useful approaches.

Method	Key Question	Primary Users	Best Suited For
Liquidation value	What would the asset fetch if sold today?	Lenders, courts	Distressed assets, loan collateral
Replacement cost	What would it cost to recreate this asset?	Insurers, engineers	Physical assets, infrastructure
Comparable value	What do similar assets sell for?	Brokers, analysts	Assets with active comparable markets

Method	Key Question	Primary Users	Best Suited For
Accounting (book) value	What does the balance sheet say?	Accountants, regulators	Financial reporting, compliance
Discounted cash flow (DCF)	What is the PV of expected future cash flows?	Corporate finance, inv. banking	Going-concern businesses, capex
Benchmark / portfolio value	What is this worth to a specific investor?	Financial engineers, portfolio mgrs.	Private or Illiquid assets, derivatives, complex instruments

The first four methods are most commonly used by accountants and appraisers. They tend to be objective, auditable, and backward-looking: they rely on observable transactions or physical measurements. The last two, DCF and benchmark valuation, are the primary tools of financial engineers. They are forward-looking, estimating value from expected future cash flows and risk, and they are the focus of this book.

It is important to recognize that **price** and **value** are not the same thing. Price is determined by supply and demand in the market. Value is the output of a model or a subjective opinion. They may be equal in an efficient market at equilibrium, but a financial engineer's job is precisely to identify and act on cases where they diverge.

1.3 Rates and Returns

1.3.1 Expected, realized, and required returns

The **realized return** on an investment is what actually happened: the cash flows received and the change in value that occurred over the holding period. It is a historical observation, known only after the fact.

The **expected return** is the investor's forecast, made at the time of investment, of what the investment will produce. Because the future is uncertain, the expected return is a probability-weighted average of all possible outcomes.

The **required return** is the minimum return the investor demands, given the investment's risk and available alternatives. It has two components: the **time value of money** (compensation for waiting) and a **risk premium** (additional compensation for bearing risk). The required return is determined by competitive market forces, not by any individual investor's preferences.

In a competitive capital market, prices adjust until the expected return on an asset equals its required return. If expected return exceeds required return, the asset is cheap. Buyers compete to purchase it, driving up its price and reducing its expected return until equality is restored. If expected return falls below required return, holders sell, pushing the price down and the ex-

pected return up. This equilibrium condition, that expected return equals required return, is the cornerstone of modern asset pricing and the foundation of the GVE introduced in Section 1.5.

1.3.2 Dollar returns and percentage returns

The **dollar return** on an investment is the change in wealth over the holding period: the cash flows received plus the change in the asset's market value. If you buy a share of stock for \$100, receive a \$3 dividend, and sell for \$110 one year later, your dollar return is \$3 + (\$110 − \$100) = \$13.

The **percentage return** divides the dollar return by the initial investment. In the example above, the percentage return is \$13 / \$100 = 13%. Percentage returns allow easy comparison across investments of different sizes, and they work well for straightforward long positions.

However, percentage returns have a critical limitation: they are undefined, or misleading, whenever the initial investment is zero or negative, or if additional calls can be made on the investor to provide capital.

1.4 When Percentage Returns Break Down

Short positions. A short seller borrows a security, sells it, and later repurchases it. When you short a stock at \$50, you receive \$50 in cash, meaning that your initial outlay is effectively negative. If the stock rises to \$80, your dollar loss is -\$30. The "percentage return" would be −\$30 / −\$50 = +60%, which is arithmetically derived from a losing trade. The positive sign is economically nonsensical. Worse, because a short position has theoretically unlimited loss potential, it is not even clear what the maximum effective investment is, making the denominator fundamentally ill-defined.

Forward contracts. A forward contract requires no initial investment: both parties agree to exchange an asset at a future date at a price set today. The initial value of the contract is zero, so any dollar return divided by zero is undefined. Yet forward contracts clearly have contingent or potential economic value. One party gains and the other loses as prices move. Dollar returns capture this; percentage returns cannot.

What "zero initial value" actually means. The phrase should be interpreted carefully. At the moment of inception, the contract is fair to both parties because the agreed price reflects every piece of information then available: today's spot price, the cost of carrying the asset to maturity, and the market's consensus about where the underlying is heading. With both parties pricing off the same information set, neither one needs to be paid to enter, so the contract changes hands at zero value. As time passes, fresh information arrives and prices move. The contract's value tracks those moves: it accrues to one side and away from the other. By the maturity date the contract is almost never worth zero. It has become an asset to whichever party guessed right and a liability to the other. The initial zero is a forecast, not a description of what the contract will be worth at any later date.

Other unfunded positions. More broadly, whenever a position requires no cash outlay or generates a net cash inflow at initiation, percentage returns are either undefined or misleading. For example, one may buy a life insurance policy at a low price not realizing how much more must be paid to keep the policy in force, meaning that returns should not be computed from the initial investment. Another example: when selling short, positions are margined, meaning that additional funds must be deposited to cover possible losses in closing the position. This amount is unknown at inception, no simple return measure will suffice.

For these reasons, this book uses dollar returns as its primary language. Percentage returns appear where they are well-defined and intuitive, particularly for long positions in stocks and bonds in later chapters. The General Valuation Equation developed in the next section is stated entirely in dollar terms.

Worked Example: Dollar vs. Percentage Returns on a Short Sale

Situation: You sell short GameStop at \$40. The price rises to \$80 one year later and you close the position.

Investment value: -\$40

Dollar return: \$40 (proceeds) − \$80 (repurchase) = −\$40 per share. You lost \$40.

Percentage return: −\$40 / −\$40 = +100%. A positive number on a losing trade!

Takeaway: Dividing a negative dollar return by a negative initial position produces a positive percentage. The dollar return correctly communicates economic reality; the percentage return misleads.

1.5 The Most Beautiful Equation in Finance

The most powerful valuation concept in finance can be stated in a single sentence: the price of an asset must be set so that the required return on the asset equals its expected return, measured in dollar terms. This is not an assumption. It is an equilibrium condition. In competitive markets, prices adjust until it holds; any deviation creates a trading opportunity that rational investors will exploit.

1.5.1 Building the required return

The required return on an asset has two components. The first is the **time value of the investment**: even a risk-free asset must offer a return equal to the risk-free periodic interest rate r, since investors have the alternative of holding risk-free securities. If the asset is currently worth V_0, the time value component in dollars is rV_0, the interest paid over one period on the cash component of the investment.

The second component is **risk compensation**: a risky asset must generally offer additional return above the risk-free rate to attract investors. We denote the market's **cost of risk per unit** by k and the dollar risk of the asset's uncertain cash flows by σ. This σ is not always the same as

the standard deviation. The total risk compensation required is $k\sigma$. In later chapters, when we use different risk measures, we simply redefine k and σ accordingly. The total required return is thus $rV_0 + k\sigma$.

1.5.2 Building the expected return

The expected return on an asset also has two components. The first is the **expected capital gain**: the expected change in the asset's value over the holding period, denoted $E[V_1] - V_0$. The second is the **expected cash flow** received during the period, denoted $E[C_1]$. The total expected return in dollars is thus $E[V_1] - V_0 + E[C_1]$.

1.5.3 The General Valuation Equation

Setting required return equal to expected return yields the GVE:

$$rV_0 + k\sigma = \mathrm{E}[V_1] - V_0 + \mathrm{E}[C_1]$$

Or, stated in words: *Time value + Risk compensation = Expected capital gain + Expected cash flow.* Note that $E[V_1]$ is computed after the dividend or cash flow is paid.

In a risk-free world ($\sigma = 0$), the GVE reduces to the familiar present value formula. With risk, the left side contains an additional term $k\sigma$ that penalizes volatile assets and can, when risk is very large, produce a negative valuation, meaning a rational investor would only accept the asset if paid to take it.

1.5.4 Notation Reference

Symbol	Definition
r	Risk-free periodic interest rate. Subscripts mark required returns on specific assets, e.g. r_e for required return on equity.
k	Cost of risk per unit of standard deviation (or other defined risk measure).
σ	Dollar standard deviation (or other risk measure) of cash flows or asset value.
$V_0,\ V_1$	Asset value at time 0 (today) and time 1 (end of period).
C_1	Cash flow received at time 1.

Worked Example: Risk-Free Valuation of a Simple Bond

A government bond promises to pay \$110 in exactly one year and makes no other payments. The risk-free periodic interest rate is r = 10%. What is the bond's value today?

Since the bond is risk-free, $\sigma = 0$ and $k\sigma = 0$. The capital gain is $110 - V_0$ (the bond pays off 110 and disappears). The remaining cash flow is zero. The GVE gives:

$$r\ V_0 = 0.10 \times V_0 = 110 - V_0 + 0$$

$$1.10 \times V_0 = 110 \Rightarrow V_0 = \$100$$

This is the familiar present value formula. The GVE contains it as a special case.

Worked Example: Risky Valuation of Alibaba Stock

You believe Alibaba will pay a $2 dividend next year and that the share price will be $128 in one year, with a dollar standard deviation of $15. The cost of risk is k = 0.60 per dollar of standard deviation. The risk-free rate is r = 10%. What is Alibaba's value today?

Apply the GVE:

$$rV_0 + k\sigma = E[V_1] - V_0 + E[C_1]$$

$$0.10\ V_0 + (0.60)(15) = (128 - V_0) + 2$$

$$0.10\ V_0 + 9 = 130 - V_0$$

$$\mathbf{1.10\ V_0 = 121 \Rightarrow V_0 = \$110}$$

Note: V_0 appears on both sides of the GVE (inside the capital gain term ($E[V_1] - V_0$). Always collect V_0 terms before solving.

The expected return of the Alibaba investment can be expressed in dollar terms or percentage terms. In dollar terms, the total expected return is $20 *(= 128 + 2 - 110)*, with $11 *(= 110 x 0.10)* for the cash component and $9 for risk compensation *(0.60)(15)*. In percentage terms, these can be divided by $110, giving a 10% return on cash and 8.18% expected risk premium.

A choice to make

As we saw in the Alibaba Worked Example, it is possible to express cash compensation and risk compensation in dollar terms or percentage terms. Does it matter? Is one better than the other?

In practice, financial analysts who value well-capitalized stocks and bonds find it convenient to express return and risk premium values in percentages. This facilitates easy comparisons between similar assets, for example, allowing one to sort stocks according to expected future percentage return or percentage risk, or to judge bonds with different credit ratings under a common standard. Since the 1960s, equity and debt models have been built using expected percentage returns, standard deviations of returns, and their correlations. Indeed, this is the foundation of Modern Portfolio Theory, as pioneered by Harry Markowitz.

Yet, modern finance requires a more general paradigm. We saw earlier in this chapter that percentage returns are inadequate to capture returns and risks of modern capital market instruments,

such as short sales, forward positions and options.

The table below demonstrates that the GVE can also be applied when the risk charge is assumed to be proportional to the asset value. Allow the letter π to represent the risk premium as a fixed percentage of value. In this case, we replace $k\sigma$ with πV_0.

Dollar GVE	Percentage GVE
Risk charge $k\sigma$	*Risk charge* $k\sigma \leftarrow \pi V_0$
$rV_0 + k\sigma = E[V_1] - V_0 + E[C_1]$	$rV_0 + \pi V_0 = E[V_1] - V_0 + E[C_1]$
$\boldsymbol{V_0 = \{E[V_1] + E[C_1] - k\sigma\}/\{1 + r\}}$	$\boldsymbol{V_0 = \{E[V_1] + E[C_1]\}/\{1 + r + \pi\}}$
Risk charge in the numerator	*Risk charge in the discount rate*
Best for short positions, underfunded positions, derivative-like positions and contracts	*Best for well-capitalized stock and bond portfolios.*

In sum, the GVE works for a broad class of assets or contracts, and the risk charge can be forced to be proportional to asset value when it is appropriate to assume that this is reasonable.

Example: The "Elon Musk Problem"

Suppose Elon is offered a deal where he can win or lose \$1B on the toss of a fair coin. We want the present value of this deal today.

If the toss is imminent, we can assume $r = 0$. Because the coin is fair, $E[C_1] = E[V_1] = 0$, and the dollar standard deviation of the outcome is exactly \$1B.

The percentage form of the GVE returns $V_0 = 0$: the deal is worth nothing. The dollar form returns $V_0 = -k\ \$1B$: a strictly negative value whenever the market charges anything for risk.

The two answers disagree, and the dollar form is right. A bank quoting this trade with $k = 25\%$ would price it at $-\$0.25B$, paying its counterparty \$0.25B to absorb the risk, because nobody, not even Elon Musk, should bear the risk of a \$1B coin toss without compensation. Note that this has nothing to do with risk aversion, and everything to do with the existence of more lucrative risks that Elon might take. Therefore, from here on we use the dollar form of the GVE as the working framework of the chapter, recognizing that the percentage form can be taken as a special case.

Personal Finance Application: Assets, Liabilities, and Your Personal Balance Sheet

Before you can apply valuation to stocks or bonds, it helps to recognize that you already have a portfolio of assets and liabilities, your personal balance sheet.

Consider a recent college graduate with the following items:

- Checking account: \$3,000 (capital asset, no risk, no capital gain)
- Used car: \$8,000 (physical asset; depreciating, an expected capital loss each year)

- Student loan: −$25,000 (liability; future cash outflows for interest and principal)
- Employment contract: the present value of expected future salary (a very large intangible asset)

Each item can, in principle, be valued using the GVE. The student loan, for example, is a contract: at issuance, it had zero net value (the borrower received cash equal to the loan amount). Over time, if interest rates fall, the loan becomes a bigger liability, and refinancing becomes valuable. This is exactly the logic of bond valuation in Chapter 4.

Takeaway: Valuation is not just for professionals. Every financial decision you make, whether to refinance a loan, buy vs. lease a car, or invest in a retirement account, is at its core a valuation problem.

1.6 Chapter Summary

Value is not a fixed property of an asset. It is the output of a model, and different models produce different values. Private valuations, comparable-based valuations, accounting valuations, and formal mathematical models are all legitimate approaches, each suited to different purposes. This book focuses on formal valuation models because they are explicit about their assumptions and therefore can be customized for any given valuation problem.

Financial instruments fall into three categories: assets (and their mirror images, liabilities), contracts, and contingent claims. All three require valuation, and all three can be analyzed using a common framework.

Returns are best measured in dollars. Percentage returns work well for long positions in well capitalized stocks and bonds. Percentage returns fail for short positions, forward contracts, and any position initiated at zero or negative cost. The dollar GVE has no such limitation.

The General Valuation Equation states that the required return, time value plus risk compensation, must equal the expected return, capital gain plus cash flows in a competitive capital market. In later chapters we will show how this single equation can be used to derive the CAPM (Capital Asset Pricing Model), Black-Scholes-Merton European option pricing model, forward/futures pricing and structured products as special cases, all of which are covered in detail in later chapters. It is the spine of this textbook.

1.7 Key Terms

Term	Definition
Asset	Any resource expected to produce positive cash flows or capital appreciation.

Term	Definition
Benchmark value	Value estimated by replicating an asset's cash flows using traded securities or contracts of known price.
Capital asset	An asset defined primarily by the cash flows it produces (e.g., a stock or bond).
Capital gain	The increase in an asset's market value over a holding period.
Contract	A bilateral agreement to exchange something at one or more future dates, typically requiring no initial investment.
Contingent claim	An instrument whose payoff depends on whether a specified event occurs (e.g., an option or insurance policy).
Cost of risk (k)	The market's price per unit of risk (standard deviation), expressed in dollar return per dollar of σ.
DCF	Discounted cash flow: a valuation method that estimates value as the present value of expected future cash flows.
Dollar return	The total change in wealth from an investment: cash flows received plus the change in asset value.
Expected return	The probability-weighted average of all possible investment outcomes, estimated at the time of investment.
GVE	General Valuation Equation: $r\,V_0 + k\sigma = E[V_1] - V_0 + E[C_1]$. Required return = expected return.
Liability	An obligation to make future payments; the mirror image of an asset.
Limited liability	The legal protection that limits an equity holder's loss to the amount invested.
Percentage return	The dollar return divided by the initial investment; undefined or misleading when the denominator is zero or negative.
r	The risk-free periodic interest rate (the default meaning of r throughout this book).
Required return	The minimum return demanded by investors, equal to time value ($r\,V_0$) plus risk compensation ($k\sigma$).
Risk compensation	The additional expected return above the risk-free rate demanded for bearing uncertainty; equals $k\sigma$ in the GVE.
Time value of money	The principle that a dollar today is worth more than a dollar in the future, because it can be invested at rate r.

1.8 Interview Questions

These questions test conceptual understanding and the ability to communicate ideas clearly and precisely. Aim for answers of two to four sentences.

1. What is the difference between valuation and a valuation model?
2. What valuation models are primarily used by accountants, and why?
3. What valuation models are primarily used by financial engineers, and why?
4. What is the difference between a financial asset and a contract? Give an example of each.
5. Why are assets and liabilities mirror images of each other?
6. What is limited liability, and why does it affect the value of equity?
7. What is the difference between a realized return, an expected return, and a required return?
8. Why must expected returns equal required returns in a competitive capital market?
9. What are the two components of the required return in the GVE?
10. When are percentage returns undefined? When are they misleading?
11. What is the difference between price and value? In a stock? In a forward contract?
12. What is the relationship between expected and required returns in an efficient market, and what mechanism enforces it?
13. Why might a private valuation differ from the market price of the same asset?
14. Name three famous financial models that are special cases of the GVE.

1.9 Numerical Problems

Difficulty: S = Straightforward | M = Moderate | C = Challenge

1. [S] Valuation Method Matching. For each situation, identify which valuation method (liquidation, replacement cost, comparable, book, DCF, or benchmark) is most appropriate, and briefly explain why.

(a) A bank is deciding how much to lend against a warehouse as collateral.

(b) An insurance adjuster is assessing the claim after a house fire.

(c) An equity analyst is publishing a buy/sell recommendation on Apple stock.

(d) A financial engineer is pricing a six-month crude oil forward contract.

(e) An acquirer is evaluating the purchase of a privately held manufacturing firm.

2. [S] Dollar and Percentage Returns. For each scenario, calculate the dollar return and the percentage return, or explain why the percentage return is undefined or misleading.

(a) You buy 100 shares of Microsoft at \$300, receive a \$10 dividend per share, and sell at \$330 one year later.

(b) You buy a call option for \$5 per share. At expiry the stock is below the strike; the option expires worthless.

(c) You short 50 shares of a stock at \$80. The stock rises to \$120 and you close the position.

(d) You enter a forward contract to buy one barrel of oil in three months at \$75. No money changes hands today. At delivery, spot oil is \$90.

3. [S] Distressed Company. A firm's assets are currently worth \$100 million. The firm has \$120 million of debt maturing in one year. Using limited-liability logic, what is the current value of the equity? What is the likely value of the debt? Discuss the role of limited liability in your answer.

4. [M] GVE Table for Tesla. You buy Tesla at \$400 per share. Your expectation at the time of purchase is that the stock will reach \$500 in one year. Tesla pays no dividends, and the risk-free rate is r = 10%. You assume the stock is risk-free for this exercise (kσ = 0). Fill in the GVE table below based on information known at the time you invest. Then, one year later, you discover the stock actually reached \$600. Recalculate the realized dollar return.

Component	Formula	Expected Value	Required Value
Time value of investment	$r \times V_0$		
Risk compensation	$k \times \sigma$		
Total required return	$r\ V_0 + k\sigma$		
Expected capital gain	$E[V_1] - V_0$		
Expected cash flow	$E[C_1]$		
Total expected return	Cap. gain + CF		

5. [M] GVE Valuation of Alibaba. You believe Alibaba will pay a \$2 dividend next year. You also expect the share price to rise to \$128, with a dollar standard deviation of σ = \$15. The cost of risk is k = 0.60 per dollar of standard deviation. The risk-free rate is r = 10%.

(a) What is your private valuation of Alibaba today?

(b) If Alibaba currently trades at \$115, what is the expected return at the market price? What does this imply about whether the stock is over- or underpriced relative to your valuation?

(c) Compute $\partial V_0/\partial \sigma$, $\partial V_0/\partial k$, $\partial V_0/\partial E[C_1]$, and $\partial V_0/\partial r$. Give the sign of each and explain the economic intuition.

6. [C] Investment Bank Contract. An investment bank is offered a contract that pays \$44 billion if a fair coin flip comes up heads and costs \$44 billion if it comes up tails. The bank's cost of risk is $k = 0.25$ per dollar of standard deviation. The risk-free rate is $r = 10\%$.

(a) What is the dollar standard deviation of the contract's payoff?

(b) What is the GVE value of the contract (V_0)? Should the bank accept the deal at zero upfront cost?

(c) What upfront payment would make the bank indifferent to accepting? Explain the sign of the payment.

(d) Why is the percentage return undefined for this contract? Why is this not just a technicality?

7. [M] Auto Insurance. You are an excellent driver and a skilled actuary. You have determined that your expected automobile accident losses next year are \$1,000 (assume paid in one year). The premium for a one-year policy is \$2,000, paid today. The risk-free rate is $r = 10\%$. The insurance company charges no risk premium on a net basis since your risk is idiosyncratic (diversifiable). Should you buy the insurance? Compute the present value of the contract to you, and discuss whether your answer changes if you are risk-averse.

1.10 Excel / Python Exercises

A. GVE Sensitivity Dashboard. Build a spreadsheet that implements the Alibaba GVE valuation (Problem 5). Use Excel's What-If (Data Table) feature to compute V_0 as a function of (i) k varying from 0 to 1.0 in steps of 0.1, and (ii) σ varying from \$5 to \$30 in steps of \$5. Produce a two-dimensional sensitivity table and a surface chart. What happens to value when $k \times \sigma > E[V_1] + E[C_1]$?

B. Personal Balance Sheet. Build a spreadsheet that computes your (or a hypothetical student's) personal balance sheet using the framework introduced in the Personal Finance Application. Include: savings accounts, a car (depreciating asset), student loans (liability), and the present value of expected salary over a 10-year horizon. Discount all cash flows at $r = 5\%$. What is the net present value of your estimated net future earnings? How sensitive is it to the assumed salary growth rate?

Chapter 2

Time Value of Money and Cash Flow Valuation

Learning Objectives

By the end of this chapter, students will be able to:

1. Apply the General Valuation Equation (GVE) to derive closed-form present value formulas for common cash flow patterns.
2. Value a single future payment, a flat perpetuity, and a flat annuity using the GVE derivation method.
3. Explain loan amortization as a direct application of the GVE, and calculate mortgage payments and outstanding balances.
4. Value exponentially growing perpetuities and annuities, and apply the r* substitution to simplify calculations.
5. Value linearly growing perpetuities and annuities using the GVE capital-gain argument.
6. Describe mean reversion intuitively and value mean-reverting perpetuities and annuities.
7. Use Excel financial functions (PV, PMT, NPER, RATE, NPV) correctly and recognize their limitations.

2.1 Introduction

Cash flows come in many forms, but most follow recognizable patterns. A mortgage requires equal monthly payments for thirty years. A company with steady revenue growth generates cash flows that rise at a constant percentage each year. An oil royalty stream is linked to a commodity price that tends to drift back toward a long-run level. In each case, the raw present value calculation, discounting each cash flow and summing, is straightforward in principle but impractical at scale. A 30-year mortgage requires 360 separate discounting operations; a financial engineer pricing millions of instruments cannot afford brute-force summation.

For this reason, we develop closed-form analytic formulas for the present values of frequently occurring cash flow patterns. These formulas are not tricks to be memorized. Each one will be derived, from scratch, using the General Valuation Equation introduced in Chapter 1. The GVE turns out to be a powerful machine for generating present value formulas with minimal algebra, much less than the traditional approach of summing geometric series. More importantly, each derivation builds intuition about why the formula takes the form it does.

Throughout this chapter the cash flows are assumed to be risk-free, so the risk term $k\sigma$ in the GVE is zero. This lets us focus entirely on the time-value structure of the problem. From the GVE with $k\sigma = 0$, the equilibrium condition simplifies to:

$$\boldsymbol{r V_0} = E[V_1] - V_0 + C_1 \Rightarrow V_0 = \boldsymbol{(E[V_1] + C_1)/(1+r)}$$

This is the risk-free present value relationship: the current value equals next period's expected value plus cash flow, discounted at the risk-free rate. We will use this repeatedly in each derivation.

2.2 Notation and the Discount Factor

The following notation is used consistently throughout this chapter. A summary also appears in the Key Terms section.

Symbol	Definition
r	Simple periodic interest rate (risk-free, default meaning of r from Chapter 1)
n	Number of periods; also the date of the final payment
C	Amount of a constant cash flow payment per period
C_i	Cash flow occurring at date i
C_1	First cash flow (at the end of period 1)
F	Face value or principal amount (used for bonds)
g	Percentage growth rate per period: $C_{i+1} = C_i(1+g)$
b	Fixed dollar increase per period: $C_{i+1} = C_i + b$
μ (mu)	Long-run mean level of a mean-reverting cash flow
κ (kappa)	Speed of mean reversion per period ($0 \leq \kappa \leq 1$)
V_i	Present value of the cash flow stream as of date i; V_0 = today's value
$V_1 - V_0$	Capital gain on the asset over one period (after C_1 has been paid)
d	Discount factor: $d = 1/(1+r)$
d^n	n-period discount factor: $d^n = 1/(1+r)^n$ = present value of \$1 paid at time n
r*	Adjusted rate for growing annuities: $r^* = (r-g)/(1+g)$
d^n*	n-period discount factor at rate r*: $d^n* = 1/(1+r^*)^n$

The discount factor $d = 1/(1+r)$ will appear in almost every formula. It is an analytic convenience: writing d^n is cleaner than writing $1/(1+r)^n$, and it makes formulas easier to remember and manipulate. Note that $d < 1$ whenever $r > 0$, so higher powers of d represent smaller present values, as expected.

Math Review: Geometric Series

Many present value formulas arise from summing a geometric series. Recall that for any number $x \neq 1$ and positive integer n:

$$1 + x + x^2 + \cdots + x^{n-1} = (1 - x^n)/(1 - x)$$

As n $\to \infty$, if |x| < 1, this sum converges to 1/(1−x). In present value work, x is typically d = 1/(1+r), so |x| < 1 whenever r > 0.

We will derive all formulas in this chapter from the GVE rather than this series, but the geometric series is useful for verifying results and for students who prefer algebraic confirmation.

2.3 Valuing a Single Future Payment

The simplest cash flow: a single risk-free payment of C at time n, with no payment before or after. What is its value today?

Applying the GVE one period at a time, and working backwards:

At date $n-1$: the value is the discounted value of C received one period later:

$$V_{n-1} = C/(1+r) = Cd$$

At date $n-2$: discount one more period:

$$V_{n-2} = V_{n-1}/(1+r) = Cd^2$$

Continuing back to time 0:

Present value of a single payment C at time n:

$$\boldsymbol{V_0 = C \cdot d^n = C \,/\, (1+r)^n}$$

This is the familiar discounting formula. Note that d^n is the present value of \$1 paid at time n, often called the **discount factor** or the **zero-coupon bond price** for maturity n. We will use this language extensively in Chapter 4.

A **zero-coupon bond** (ZCB) is the simplest fixed-income instrument. It pays no coupons during its life: the holder receives a single fixed payment, called the *face value* F, at maturity, and nothing else. The buyer's return in dollars equals the difference between the face value paid at maturity and the current price of the ZCB.

A small numerical example fixes the idea. Suppose a 5-year Treasury STRIP promises to pay \$1,000 in five years. With a flat risk-free rate of $r = 5\%$, the GVE with no risk premium gives $V_0 = \$1{,}000 \times d^5 = \$1{,}000 \,/\, (1.05)^5 \approx \$783.53$. That \$783.53 is exactly "the 5-year discount factor times \$1,000." Every ZCB price is a discount factor multiplied by a face value, which is why d^n is sometimes called the zero-coupon bond price for maturity n: the math is identical.

Worked Example: Valuing a Lump-Sum Inheritance

Your grandmother promises to give you \$50,000 in 10 years. The risk-free rate is r = 5% per year. What is the gift worth today?

$$V_0 = 50{,}000 \,/\, (1.05)^{10} = 50{,}000 \times d^{10}$$

$$d^{10} = 1 / (1.05)^{10} = 0.6139$$

$$\boldsymbol{V_0 = \$30{,}695}$$

Interpretation: at a 5% annual discount rate, $30,695 today is exactly equivalent to $50,000 in ten years. If the gift were offered today, $30,695 would be the fair exchange.

2.4 The Flat Perpetuity

A flat perpetuity pays a constant amount C at the end of every period, forever. Examples include preferred stock dividends, consol bonds (issued historically by the UK government), and interest-only withdrawals from a savings account.

To find V_0, we use the GVE's capital gain insight. After the first payment $C_1 = C$ is made, the remaining cash flow stream, C per period forever, is identical to the original stream. Therefore:

$$V_1 = V_0$$

The capital gain is zero: $V_1 - V_0 = 0$. Substituting into the risk-free GVE:

$$rV_0 = (V_1 - V_0) + C = 0 + C$$

Present value of a flat perpetuity (C per period, first payment in one period):

$$\boldsymbol{V_0 = C/r}$$

This is one of the most elegant results in finance. The value of a perpetuity is simply the annual payment divided by the interest rate. Notice that it follows directly from the GVE without any summation: once we recognize that $V_1 = V_0$, the algebra is immediate.

Personal Finance Application: Preferred Stock and the Perpetuity Formula

A corporation issues preferred stock that pays a fixed quarterly dividend of $1.25 per share, with no maturity date. If the required return on preferred stock is $r_p = 2\%$ per quarter (8% annually), what is the fair value of one share?

$$V_0 = C/r = \$1.25 / 0.02 = \$62.50 \text{ per share}$$

If the market price is $58, the stock's expected return exceeds the required return: it appears underpriced. If it trades at $70, it appears overpriced, the same GVE logic from Chapter 1 applies.

2.5 The Deferred Perpetuity and the Flat Annuity

2.5.1 Deferred perpetuity

A deferred perpetuity is a perpetuity whose first payment occurs not in one period but in j+1 periods. Since the perpetuity begins at date j, its value at date j is C/r. To find its present value today, we discount j periods:

$$V_0 \text{ (deferred by j periods)} = (C/r) \cdot d^j$$

2.5.2 The flat annuity as perpetuity minus deferred perpetuity

A flat annuity pays C per period for exactly n periods. The key insight is:

Annuity = Perpetuity starting now − Perpetuity starting at period n+1

The first perpetuity has value C/r today. The second perpetuity starts at date n, so its value at date n is C/r, and its present value today is $(C/r) \cdot d^n$. Therefore:

Present value of a flat annuity (C per period for n periods):

$$V_0 = (C/r)(1 - d^n)$$

Note: when $r = 0$, the formula gives $V_0 = nC$ (just sum the undiscounted payments). When $n \to \infty$, $d^n \to 0$ and the annuity reduces to the perpetuity C/r.

2.5.3 GVE verification of the annuity formula

We can also derive the annuity formula directly from the GVE. After one period, the remaining stream is an annuity of n−1 periods:

$$V_1 = (C/r)(1 - d^{n-1})$$

$$V_1 - V_0 = (C/r)(d^{n-1} - d^{n-1}(1+r)) \ldots$$

Substituting into $rV_0 = (V_1 - V_0) + C$ and simplifying recovers the same formula, confirming consistency.

Worked Example: Solving for the Mortgage Payment

A homeowner borrows **$400,000** at a fixed annual rate of **r = 6%** (0.5% per month) for **30 years** ($n = 360$ monthly payments). What is the monthly payment C?

Rearrange the annuity formula to solve for C:

$$V_0 = (C/r)(1 - d^n) \Rightarrow C = rV_0 \,/\, (1 - d^n)$$

Compute $d^n = 1/(1.005)^{360} = 0.16604$:

$$C = (0.005 \times 400{,}000) \,/\, (1 - 0.16604) = 2{,}000 \,/\, 0.83396$$

$$\boldsymbol{C = \$2{,}398.20 \text{ per month}}$$

Excel check: =PMT(0.005, 360, −400000) = \$2,398.20. The Excel PMT function is simply the annuity formula solved for C.

2.5.4 Outstanding loan balance

After making i payments, the outstanding balance is the present value of the remaining n−i payments, discounted at r:

$$\boldsymbol{V_i = (C/r)(1 - d^{n-i})}$$

This also follows from the GVE: the balance after any payment is simply the present value of the remaining annuity.

2.6 Loan Amortization: The GVE in Action

Loan amortization is the process by which a loan balance is gradually paid down over time. A standard amortizing loan has constant payments C, but each payment is split between interest and principal reduction. The GVE makes this transparent.

From the GVE for a risk-free asset with known cash flow:

$$rV_i = (V_{i+1} - V_i) + C$$

Rearranging: the new balance $V_{i+1} = V_i + rV_i - C = V_i(1+r) - C$. Each payment C covers the interest charge rV_i, with the remainder $C - rV_i$ reducing the principal.

Period i	Beginning V_i	Interest rV_i	Payment C	Principal C - rV_i	Ending V_{i+1}
1	V_0	rV_0	C	$C - rV_0$	$V_0(1+r)-C$
2	$V_0(1+r)-C$	$r[V_0(1+r)-C]$	C	C − r[...]	$V_0(1+r)^2-2C(1+r)-C$
⋮	⋮	⋮	⋮	⋮	⋮
n	(last payment covers balance)	rV_{n-1}	C	$=V_{n-1}$	0

Early in the loan, most of C covers interest. Over time, as the balance falls, the interest component shrinks and the principal reduction grows. This is why the outstanding balance on a mortgage falls slowly at first and rapidly near maturity, a fact with important implications for **refinancing decisions**. This pattern is shown in the graph below, assuming the loan is not prepaid, i.e. paid before the stated maturity date.

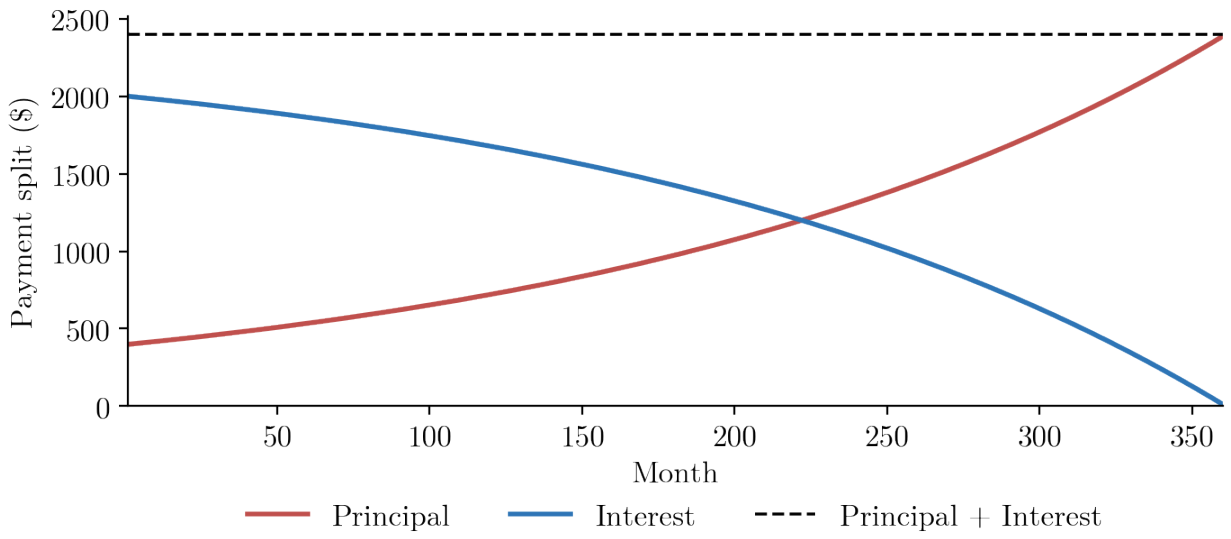


Figure 2.1: *Outstanding balance on a 30-year, $400,000 mortgage at 6% over the life of the loan, assuming no prepayment.*

Personal Finance Application: Should You Refinance Your Mortgage?

You took out a 30-year $400,000 mortgage at 6% five years ago. Monthly payment = $2,398.20 (from Worked Example above). After 60 payments, your outstanding balance is:

$$V_{60} = (C/r)(1 - d^{n-i}) = (2398.20/0.005)(1 - 1/(1.005)^{300}) = \$372{,}217$$

You are offered a new 25-year mortgage at 4.5% (r = 0.375%/month). New payment:

$$C_{new} = r \times V_{60} \,/\, (1 - d^{n-230}) = 0.00375 \times 372{,}217 \,/\, (1 - 1/(1.00375)^{300}) = \$2{,}069$$

Monthly savings: $2,398 − $2,069 = $329. If closing costs are $6,000, breakeven ≈ 18 months. Whether to refinance depends on how long you plan to stay in the home, a present value problem we return to in the exercises.

2.7 The Growing Perpetuity and the Growing Annuity

2.7.1 Growing perpetuity

A growing perpetuity pays C_1 at the end of period 1, then grows at rate g per period forever: $C_i = C_1(1+g)^{i-1}$. This pattern describes a firm's dividends under the Gordon Growth Model, or a salary growing steadily with inflation.

After the first payment, the remaining stream is also a growing perpetuity, but with a first payment of $C_1(1+g)$. Therefore $V_1 = V_0(1+g)$, and the capital gain is:

$$V_1 - V_0 = gV_0$$

Substituting into the GVE:

$$rV_0 = gV_0 + C_1 \Rightarrow (r-g)V_0 = C_1$$

Present value of a growing perpetuity (first payment C_1, growth rate g, r > g):
$$V_0 = C_1 \,/\, (r - g)$$

The condition $r > g$ is required for convergence: if cash flows grow faster than the discount rate forever, the present value is infinite. This is the **Gordon Growth Model** (also called the Dividend Discount Model), one of the earliest and most widely used equity valuation formulas.

Corporate Finance Application: Gordon Growth Model for Equity

A mature company is expected to pay a dividend of \$3.00 per share next year, growing at g = 4% per year indefinitely. The required return on equity is $r_e = 10\%$. What is the fair value of a share?

$$V_0 = C_1 \,/\, (r_e - g) = \$3.00 \,/\, (0.10 - 0.04) = \$50.00$$

If the stock trades at \$45, its expected return at that price is $\$3/(45-4) + 0.04 \approx 0.107$, which exceeds $r_e = 10\%$, suggesting it is underpriced relative to this model.

Sensitivity: A 1 percentage point drop in the required return (to 9%) raises value to $\$3/(0.09-0.04) = \$60$, a 20% increase. Growing perpetuity values are highly sensitive to r−g near zero, a key limitation of the model.

2.7.2 The r* trick: growing annuity as a flat annuity

A growing annuity pays C_1 in period 1, growing at g per period for n periods. It can be valued using a neat substitution. Define:

$$r^* = (r - g) \,/\, (1 + g),\ C_0 = C_1 \,/\, (1+g),\ d^n{}^* = 1/(1+r^*)^n$$

Then the growing annuity has the same value as a flat annuity with payment C_0 and rate r*:

$$V_0 = (C_0/r^*)(1 - d^n{}^*) \text{ [for } r \neq g\text{]}$$

When r = g: $V_0 = nC_1$ (undiscounted sum)

Intuitively, dividing each payment by $(1+g)_i$ deflates it to a constant C_0, and dividing by $1+g$ in the discount rate compensates for this deflation. This substitution transforms any growing annuity problem into a flat annuity problem, which can then be solved in Excel with =PV(r*, n, C_0).

Worked Example: Salary Annuity with Raises

You are negotiating a 5-year employment contract. Your starting salary is \$80,000 per year, paid at year end, with guaranteed 3% annual raises. The discount rate is r = 7%. What is the

present value of this compensation stream?

Compute: $r^* = (0.07-0.03)/(1.03) = 0.03883$, $C_0 = 80{,}000/1.03 = \$77{,}670$

$$d_5{}^* = 1/(1.03883)^5 = 0.8265$$

$$V_0 = (77{,}670/0.03883)(1-0.8265) = \$2{,}000{,}000 \times 0.1735 = \$346{,}907$$

For comparison, the undiscounted sum of payments is $\$80{,}000 \times [(1.03)^5-1]/0.03 = \$424{,}646$. The difference ($77,739) is the cost of waiting to receive those cash flows.

2.8 Linear Growth Annuities and Perpetuities

Not all cash flow growth is exponential. In some applications, particularly those linked to arithmetic random walks, payments increase by a fixed dollar amount b each period rather than a fixed percentage. For instance, a step-down bonus plan might pay $100,000 in year 1, $90,000 in year 2, $80,000 in year 3, and so on.

To build intuition, start with the canonical **linear growth perpetuity**: payments of $1, $2, $3, ... starting one period from now.

$$V_0 = PV\{1,\ 2,\ 3,\ \ldots\}$$

After one payment, the remaining stream is $2, $3, $4, ..., which is just $1, $2, $3, ... plus a flat perpetuity of $1/period. Therefore:

$$V_1 = PV\{2,3,4,\ldots\} = V_0 + PV\{1,1,1,\ldots\} = V_0 + 1/r$$

$$V_1 - V_0 = 1/r$$

Substitute into the GVE (with $C_1 = 1$):

$$rV_0 = 1/r + 1 \Rightarrow V_0 = (1+r)/r^2$$

Linear growth perpetuity PV{1, 2, 3, ...}:

$$\boldsymbol{V_0 = (1+r) \,/\, r^2}$$

General case PV{b, 2b, 3b, ...}: multiply by b

2.8.1 Linear growth annuity

For a finite annuity $PV\{1,\ 2,\ 3,\ \ldots,\ n\}$, the same GVE argument applies but the last payment matters. After one period:

$$V_1 = PV\{2,3,\ldots,n,0\}$$

$$V_1 - V_0 = PV\{1,1,\ldots,1,-n\} = (1/r)(1-d^{n-1}) - nd^n$$

Substituting and simplifying (using $d^n(1+r) = d^{n-1}$):

Linear growth annuity PV{1, 2, 3, . . . , n}:

$$V_0 = [(1+r)/r^2](1 - d^n) - (n/r)d^n$$

General case: multiply by b (the fixed dollar increment per period)

Any other linear growth pattern $PV\{a, a+b, a+2b, \ldots\}$ can be decomposed using linearity: $V_0 = PV\{\text{flat annuity of } a\} + b \times PV\{1,2,\ldots,n\}$. This decomposition is powerful: it lets us value any linearly growing stream using formulas we already know.

Worked Example: Step-Down Bonus Plan

A broker receives deferred compensation: \$100,000 in year 1, decreasing by \$10,000 per year for 10 years (\$100K, \$90K, \$80K, . . . , \$10K). The discount rate is r = 8%. What is the present value of the total compensation?

Decompose: the stream is $PV\{100,90,80,\ldots,10\}$ (in thousands).

This equals $110 \times PV\{1,1,\ldots,1\} - 10 \times PV\{1,2,\ldots,10\}$ (all in \$K).

Flat annuity of \$110K for 10 years:

$$V_0 = (110/0.08)(1-1/(1.08)^{10}) = 1375 \times 0.5368 = \$738.1K$$

Linear growth annuity $PV\{1,2,\ldots,10\}$

$$= [(1.08)/0.0064](1-0.4632) - (10/0.08)(0.4632)$$

$$= 168.75 \times 0.5368 - 57.9 = 90.6 - 57.9 = 32.7$$

Present value of bonus stream: $\boldsymbol{V_0 = 738.1 - 10 \times 32.7 = \$411.1K}$

2.9 Mean-Reverting Cash Flows

2.9.1 What is mean reversion?

Some financial variables, such as interest rates, commodity prices, agricultural prices, profit margins, do not grow without bound in either direction. Instead they tend to drift back toward a long-run equilibrium level. This behavior is called mean reversion. When oil prices are high, increased supply and reduced demand push them back down; when they are low, the opposite forces push them up.

Formally, a **mean-reverting (deterministic) process** evolves as:

$$S_{t+1} = S_t + \kappa(\mu - S_t)$$

where μ = long-run mean, κ = speed of mean reversion ($0 \leq \kappa \leq 1$)

When $S_t > \mu$, the term $\kappa(\mu - S_t)$ is negative and the process drifts downward; when $S_t < \mu$, it drifts upward. The speed $\kappa = 0$ gives no mean reversion (a random walk); $\kappa = 1$ gives complete

mean reversion in one period (immediate return to μ).

Here is an illustration of mean-reverting oil price projections given several different starting points, but the same speed of mean reversion and the same long-run mean, which is shown with the dashed line.

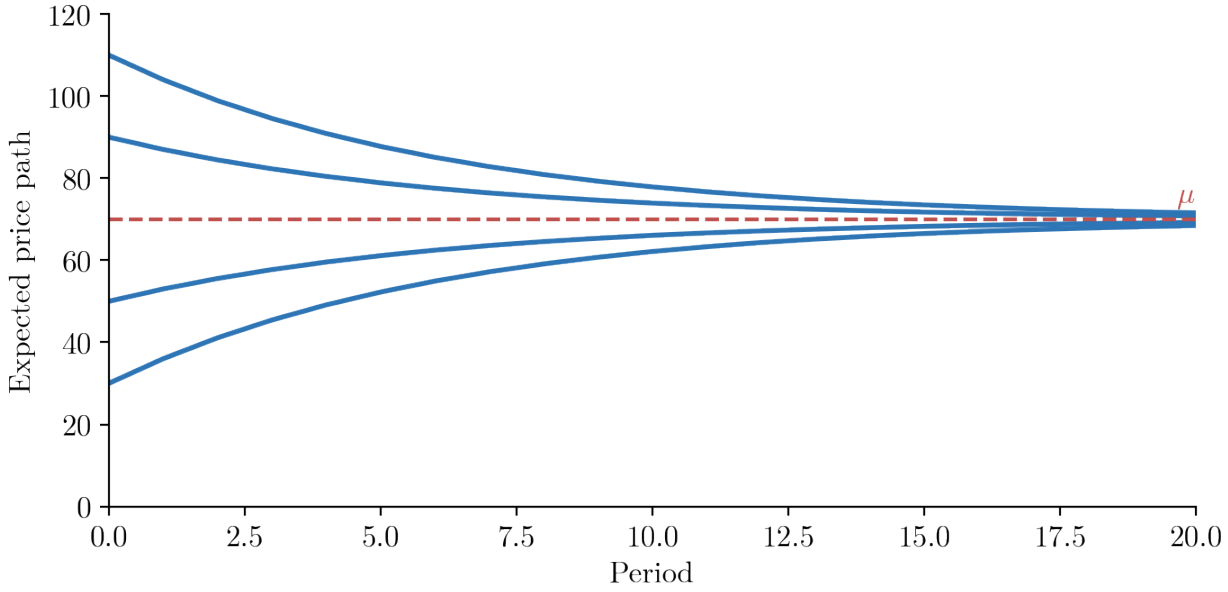


Figure 2.2: *Mean-reverting oil price projections from several different starting points, sharing the same long-run mean μ (dashed line) and speed of mean reversion κ.*

2.9.2 Valuing a mean-reverting perpetuity

Suppose a cash flow follows $S_{t+1} = S_t + \kappa(\mu - S_t)$ with S_0 today. We want $V_0 = PV\{S_1, S_2, S_3, \ldots\}$

The GVE capital gain argument:

$$V_1 - V_0 = PV\{S_2 - S_1, S_3 - S_2, \ldots\} = PV\{\kappa(\mu - S_1), \kappa(\mu - S_2), \ldots\}$$

$$= \kappa \times PV\{\mu, \mu, \mu, \ldots\} - \kappa \times PV\{S_1, S_2, \ldots\} = \kappa\mu/r - \kappa V_0$$

Substituting into the GVE:

$$rV_0 = \kappa\mu/r - \kappa V_0 + S_1 \Rightarrow V_0(r+\kappa) = \kappa\mu/r + S_1$$

After substituting $S_1 = S_0 + \kappa(\mu - S_0) = S_0(1-\kappa) + \kappa\mu$ and simplifying:

Present value of a mean-reverting perpetuity:
$$\boldsymbol{V_0 = \mu/r + (S_0 - \mu)/r^* \text{ where } r^* = (r+\kappa)/(1-\kappa)}$$

Interpretation: the value decomposes into two parts. The first term, μ/r, is the present value of a flat perpetuity at the long-run mean. The second term is a growing perpetuity with initial

value $S_0 - \mu$ (the deviation from mean) growing at rate $g = -\kappa$ (i.e., shrinking by κ per period). The second term has the right sign: if $S_0 > \mu$, the second term is positive, adding to value because early cash flows are above average.

2.9.3 Mean-reverting annuity

For a finite annuity of n periods, we simply apply the perpetuity-minus-deferred-perpetuity technique to each component:

Present value of a mean-reverting annuity (n periods):

$$V_0 = (\mu/r)(1 - d^n) + [(S_0 - \mu)/r^*](1 - d^{n*})$$

where $r^* = (r+\kappa)/(1-\kappa)$ and $d^{n*} = 1/(1+r^*)^n$

Worked Example: Oil Royalty Stream

An oil royalty pays \$50/barrel × (production of 1,000 barrels/year) = \$50,000/year. Current oil price is S_0 = \$50/bbl; long-run mean μ = \$70/bbl; mean reversion speed κ = 0.20/year. Discount rate r = 8%. The royalty lasts 20 years. What is the present value?

Compute: $r^* = (0.08+0.20)/(1-0.20) = 0.28/0.80 = 0.35$

$$d^{20} = 1/(1.08)^{20} = 0.2145 \quad d^{20*} = 1/(1.35)^{20} = 0.00186$$

First term (flat annuity at long-run mean): $(70{,}000/0.08)(1-0.2145) = \$875{,}000 \times 0.7855 = \$687{,}300$

Second term (mean-reversion correction, $S_0 - \mu = -20$): $(-20{,}000/0.35)(1-0.00186) = -57{,}143 \times 0.9981 = -\$57{,}035$

$$V_0 = \$687{,}300 - \$57{,}035 = \$630{,}265$$

Intuition: because current oil is below its long-run mean, early cash flows are lower than average. The correction term is negative, reducing present value relative to the case where oil stayed at its mean throughout.

2.10 Excel in Practice: Financial Functions

Excel contains built-in functions that implement the flat annuity formula and its inverses. These functions are simply the closed-form results from Section 2.5, packaged for convenience. Understanding the underlying formulas prevents errors and allows you to handle cases Excel cannot, such as growing or mean-reverting cash flows.

Function	Arguments	Purpose	Book equivalent
=PV(rate,nper,pmt,[fv],[type])	rate=r, nper=n, pmt=C	Present value of a flat annuity	$V_0 = (C/r)(1-d^n)$
=PMT(rate,nper,pv,[fv],[type])	pv=V_0	Payment on a flat annuity	$C = rV_0/(1-d^n)$
=NPER(rate,pmt,pv,[fv],[type])	pmt=C, pv=V_0	Number of periods to pay off V_0	Solve annuity formula for n
=RATE(nper,pmt,pv,[fv],[type],[guess])	nper=n, pmt=C	Periodic interest rate	Solve annuity formula for r
=NPV(rate,value1,[value2],...)	value1=C_1 (first payment in period 1)	PV of arbitrary cash flow stream	$V_0 = \Sigma\ C_i d_i$
=XNPV(rate,values,dates)	dates=Excel date values	PV with irregular timing	V_0 on exact dates
=IRR(values,[guess])	values=cash flows starting at t=0	Rate making NPV=0	Solve $\Sigma C_i d_i = 0$ for r

Critical tips for Excel financial functions:

- All parameters must be in the same time units. If payments are monthly, enter the monthly rate (annual rate ÷ 12) and number of months (years × 12).
- Enter PV as a negative number when PMT is positive, and vice versa. Excel treats cash outflows as negative.
- The [type] argument: 0 (default) = payments at end of period (ordinary annuity); 1 = payments at beginning of period (annuity due). Most financial problems use type=0.
- NPV in Excel assumes the first value is in period 1 (not period 0). If you have a time-0 cash flow (e.g., an initial investment), add it separately: =initial + NPV(rate, future_flows).
- For growing, linear, or mean-reverting annuities, use the formulas from this chapter directly, or build an explicit cash flow column and use =NPV.

2.11 Chapter Summary and Present Value Reference Table

The following table summarizes every cash flow type valued in this chapter. Keep it as a reference throughout the course. It will be used repeatedly in Chapters 3 through 12.

Cash flow pattern	First payment	Present value V_0	Condition
Single payment C at time n	n periods	$C \cdot d^n = C/(1+r)^n$	$r \geq 0$
Flat perpetuity	1 period	C/r	$r > 0$
Flat annuity (n periods)	1 period	$(C/r)(1-d^n)$	$r > 0$ (if r=0: nC)
Deferred flat perpetuity	j+1 periods	$(C/r) \cdot d^j$	$r > 0$
Growing perpetuity (growth g)	1 period	$C_1/(r-g)$	$r > g$
Growing annuity (growth g)	1 period	$(C_0/r^*)(1-d^{n*})$	$r^*=(r-g)/(1+g)$; $r \neq g$
Linear growth perpetuity {1,2,3,...}	1 period	$(1+r)/r^2$	$r > 0$
Linear growth annuity {1,2,...,n}	1 period	$[(1+r)/r^2](1-d^n) - (n/r)d^n$	$r > 0$
Mean-reverting perpetuity	1 period	$\mu/r + (S_0-\mu)/r^*$ $[r^*=(r+\kappa)/(1-\kappa)]$	$r>0$, $0<\kappa<1$
Mean-reverting annuity (n periods)	1 period	$(\mu/r)(1-d^n) +$ $[(S_0-\mu)/r^*](1-d^{n*})$	same

2.12 Key Terms

Term	Definition
Annuity	A finite sequence of equally spaced payments of equal or growing amounts.
Annuity due	An annuity whose first payment occurs immediately (at time 0) rather than one period from now.
Amortization	The gradual repayment of a loan principal through periodic payments that cover both interest and principal reduction.
d (discount factor)	$d = 1/(1+r)$; the present value of \$1 paid one period from now.

Term	Definition
d^n	The n-period discount factor: $d^n = 1/(1+r)^n$; the present value of \$1 paid at time n.
Deferred annuity	An annuity whose first payment occurs more than one period from now.
Gordon Growth Model	The present value of a growing perpetuity: $V_0 = C_1/(r-g)$. Widely used for equity valuation.
Linear growth annuity	An annuity whose payments increase by a fixed dollar amount each period.
Mean reversion	A process in which the value tends to drift toward a long-run mean μ at speed κ per period.
Ordinary annuity	An annuity with first payment one period from now (the default assumption in this book).
Perpetuity	An annuity with an infinite number of payments.
r^* (adjusted rate)	$r^* = (r-g)/(1+g)$; the discount rate that transforms a growing annuity into an equivalent flat annuity.
S_0, μ, κ	Starting value, long-run mean, and speed of mean reversion in a mean-reverting cash flow model.

2.13 Interview Questions

These questions test conceptual understanding. Aim for 2 to 4 sentence answers.

1. Why do financial engineers need closed-form present value formulas rather than just summing cash flows in a spreadsheet?

2. Explain intuitively why the present value of a flat perpetuity is C/r. What happens to value as r approaches zero? Why?

3. What is the key GVE insight used to derive the perpetuity formula? How does the same insight generalize to the growing perpetuity?

4. A flat annuity can be viewed as the difference between two perpetuities. Explain this in your own words.

5. Explain the amortization process. Why does the principal reduction in an early loan payment differ from that in a later payment?

6. What is the Gordon Growth Model, and what are its key limitations?

7. What does the r* substitution accomplish? When would you use it?

8. What is mean reversion, and why does it occur in some financial variables but not others (e.g., interest rates vs. stock prices)?

9. In the mean-reverting perpetuity formula, what does each of the two terms represent?

10. Why is the condition r > g necessary for the growing perpetuity formula to produce a finite value?

11. An investor says: "The =PV function in Excel gives me everything I need for cash flow valuation." When is this statement true, and when does it fail?

12. How does the outstanding balance on a loan change over time, and what is the GVE explanation for this?

2.14 Numerical Problems

Difficulty: S = Straightforward | M = Moderate | C = Challenge

1. [S] Discount Factors. Compute the discount factor d^n for each of the following:

(a) r = 5%, n = 1, 5, 10, 20, 30

(b) r = 10%, n = 1, 5, 10, 20

(c) r = 0%, n = 10

What general pattern do you observe as n increases? As r increases? What does a very small d^n mean economically?

2. [S] Perpetuity Valuation. Value each of the following perpetuities at r = 8%:

(a) A preferred stock paying \$5/share per year forever.

(b) A consol bond paying \$40/year forever.

(c) A property generating \$20,000/year in net rental income forever.

(d) For each, what would happen to the value if r rose to 10%? Calculate the percentage change.

3. [S] Mortgage Payments. Compute the monthly payment for the following mortgages (all have monthly payments):

(a) \$300,000 loan, 6% annual rate, 30 years.

(b) \$300,000 loan, 4.5% annual rate, 15 years.

(c) \$500,000 loan, 7% annual rate, 30 years.

(d) For loans (a) and (b), how much of the first payment is interest? Principal? What is the outstanding balance after 5 years?

4. [M] Government Bond. A government bond pays an annual coupon of \$60 and repays \$1,000 face value in 8 years. The discount rate is 4% per year.

(a) What is the bond's present value? (Hint: separate the coupon stream and the face value payment.)

(b) If the discount rate rises to 6%, what is the new price? What is the percentage change?

(c) Confirm your answer using the formula $V_0 = (C/r)(1-d^n) + F \cdot d^n$.

5. [M] Auto Accident Settlement. Your client was awarded a settlement of \$50,000 next year, growing at 2% per year for a total of 25 annual payments. The growth is meant to compensate for wage inflation. Her discount rate is 8%.

(a) What is the present value of the settlement?

(b) The insurance company offers to double each payment (\$100,000/year) for a shorter period, keeping the same total present value. For how many years can she receive \$100,000/year?

(c) What discount rate would make the 25-payment settlement equivalent to a 10-payment settlement of \$60,000/year?

6. [M] Oil Well Valuation. An oil well produces 500,000 barrels/year for the first 10 years, then 250,000 barrels/year for the following 10 years. Current oil price $S_0 = \$60$/bbl, long-run mean $\mu = \$80$/bbl, mean reversion speed $\kappa = 0.15$/year. Fixed costs = \$3,000,000/year; variable costs = \$12/barrel. Discount rate = 9%.

(a) What is the expected net cash flow in years 1, 5, 10, and 15 (deterministic, using the mean-reverting formula for S_t)?

(b) Value the 10-year high-production phase using the mean-reverting annuity formula.

(c) Value the 10-year low-production phase as a deferred mean-reverting annuity.

(d) What is the total present value of the oil well? Check your answer in Excel by computing each year's cash flow and summing.

7. [M] Retirement Planning. You plan to retire in 35 years and want to receive \$80,000 per year (in today's dollars) for 25 years of retirement. Inflation is expected to be 3% per year. The nominal discount rate is 7%.

(a) What nominal dollar amount will you need as your first retirement payment (in 35 years)?

(b) What is the present value, at retirement, of the 25-year retirement income stream (growing at 3%/year in nominal terms)?

(c) How much must you save per year (equal nominal payments) over the 35 years to fund this retirement nest egg?

(d) If you already have $50,000 saved, how does that change the required annual savings?

8. [C] Marketing Program Value. A new B2B marketing program is expected to sign one major new corporate client per year for the next 15 years. Each client generates $50,000 in revenue per year in perpetuity, starting one year after they sign. Signing costs $20,000 per client, paid at signing. The corporate discount rate is 12%.

(a) What is the present value of revenue from a client signed in year i?

(b) Using the linear growth annuity or an alternative approach, find the total present value of all 15 clients' revenue streams.

(c) Subtract the present value of signing costs. What is the net present value of the marketing program?

(d) How sensitive is the answer to the assumed perpetuity growth rate? Re-solve assuming each client's revenue grows at 2%/year.

9. [C] Duration as a Linear Growth Annuity. Macaulay's duration is defined as the weighted average time of a bond's cash flows, with weights equal to each cash flow's share of total value:

$$D = [\Sigma_i \; i \cdot C_i \cdot d_i] \; / \; V_0$$

(a) Show that the numerator of D is the present value of a linearly growing annuity, specifically PV$\{C_1, 2C_2, 3C_3, \ldots, nC_n\}$.

(b) For a flat coupon bond with coupon C, maturity n, and face value F, express duration using the linear growth annuity formula from Section 2.8.

(c) Compute the duration of the bond in Problem 4(a) (C=$60, n=8, F=$1000, r=4%). Interpret the result.

(d) Modified duration = D/(1+r). Convexity involves the quadratic growth annuity (covered in the Challenge Problems). For now, compute the approximate price change $\Delta V_0 \approx -D_{\text{mod}} \times V_0 \times \Delta r$ for a 1% rise in rates.

2.15 Excel / Python Exercises

A. Bond Pricing Reference Table. Build a spreadsheet that generates a bond pricing table. The user selects a bond term (in years) in one cell. The table shows coupon rates from 0% to 10% in 1% increments across columns, and discount rates from 0% to 10% in 25 basis point increments across rows. Each cell of the table shows the bond price using the formula $V_0 = (C/r)(1-d^n) + F \cdot d^n$ with F = \$1,000. Answer: (i) How does price vary with discount rate for a fixed coupon? (ii) Use conditional formatting to highlight bonds trading at a premium vs. discount to par.

B. Retirement Savings Planner. Build a flexible spreadsheet that solves Problem 7 for arbitrary inputs. Allow the user to change: starting savings, annual contribution, pre-retirement investment return, retirement duration, inflation rate, and desired real retirement income. Display: (i) required annual savings; (ii) projected savings balance year-by-year; (iii) projected retirement income year-by-year in both nominal and real dollars. Add a sensitivity table showing required savings as a function of assumed investment return and retirement duration.

C. Oil Well Valuation with Mean Reversion. Build a spreadsheet that implements Problem 6. Part 1: compute the expected oil price path S_t for t = 1 to 20 using the mean-reversion formula, and plot it. Part 2: compute each year's net cash flow (production × (price − variable cost) − fixed cost) and use =NPV to compute present value. Part 3: use the closed-form mean-reverting annuity formula and verify it matches. Part 4: build a two-way sensitivity table of total PV vs. S_0 (40 to 80) and κ (0.05 to 0.40). What combination of low S_0 and low κ produces the lowest valuation?

Mini-Project: The Two-Loan Comparison

Objective: Design a professional Excel model comparing two loan products for a client. This project reinforces loan amortization, present value, and the economics of loan fees. It also develops Excel formatting and charting skills.

Background. A borrower needs \$5,000 for 2 years, making monthly payments. Two loans are available:

- Chase Bank: 10% simple annual rate (0.833% monthly), no fees.
- Wells Fargo Bank: 7% simple annual rate (0.583% monthly), but with a 2% origination fee (\$100 on a \$5,000 loan). The fee is financed into the loan, so the borrower receives \$5,000 but owes \$5,100 from day one.

Deliverables (all in one Excel workbook with clearly labeled sheets):

(a) [Sheet 1: Amortization] For each loan, build a full 24-row amortization table showing: period, beginning balance V_i, interest rV_i, payment C, principal reduction $C - rV_i$, and ending balance V_{i+1}. Final balance must equal exactly \$0.00.

(b) [Sheet 1: Charts] Produce a stacked bar chart showing the interest vs. principal split for each payment under each loan. Produce a line chart showing the outstanding balance under each loan over time.

(c) [Sheet 2: Payoff Analysis] For each possible early payoff date (months 1 to 24), compute the payoff balance and the present value of all remaining payments, discounted at the Chase loan rate. Plot both as functions of the payoff date. What does the relationship between payoff balance and PV of remaining payments tell you?

(d) [Sheet 3: Bank's View] Wells Fargo funds the loan at a cost of 2%/year (0.167%/month). Borrowers default at 6%/year (0.5%/month), recovering nothing. Borrowers prepay at 1%/month (making extra principal payments). Model a portfolio of 1,000 identical loans. Compute: (i) projected monthly principal and interest aggregate; (ii) PV of the loan portfolio to Wells Fargo as a function of contract rate (5 to 12 percent annual) and fee (0 to 4 percent). Present as a two-way data table.

(e) [Executive Summary: 1 page max] State which loan the borrower should choose and why, with supporting numbers. Address: total interest paid, monthly payment comparison, and effective APR of each loan.

Formatting requirements: Label all charts and column headings. Dollar values to two decimal places. Interest rates to the basis point. No extraneous data visible on the summary sheet.